

Sam Tran, Software Engineering

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PROFILE

Software Engineering student with hands-on experience building and deploying full-stack applications and automation tools. Experienced with Python, Flask, React, SQL databases, and cloud deployment on Azure. Strong focus on API integration, testing, and delivering working systems in team environments.

EDUCATION

BEng, Bachelor of Engineering: Computer Science, Software Engineering

Predicted Second-class honours, upper division (2:1 Hons)

Master of Science Cybersecurity and Artificial Intelligence,
University of Sheffield

Sheffield

EXPERIENCE

Jan 2025 — Mar 2025

Full-stack Collaboration Project, Elanco

Sheffield

- Contributed to a full-stack web application using Flask (REST APIs) and React.
- Developed Python automation scripts integrating AI models and APIs.
- Deployed applications to Azure, supporting scalability and reliability.
- Worked in an Agile team (sprint planning, stand-ups, retrospectives).
- Wrote unit tests using unittest to reduce regressions.

Mar 2023 — Aug 2023

Software Intern, Pro Check-Up

London

- Built a Python-based search system to improve database query workflows.
- Profiled and debugged performance bottlenecks using built-in tools.
- Automated data extraction with Selenium.
- Designed and optimized PostgreSQL schemas for efficient querying.
- Provided technical support and troubleshooting for internal systems.

PROJECTS

Web-based Inventory Management System

- Full-stack system using Laravel, PHP, JavaScript, and SQL.
- Implemented backend models, controllers, and database schemas.
- Built a responsive UI with Bootstrap.
- Wrote PHPUnit tests to reduce post-deployment bugs.

Diet and Recipe Manager – WinForms C# Application

- C# WinForms application for managing recipes and meal plans.
- Applied OOP principles for maintainable code.
- Designed SQL Server schemas for complex queries.
- Implemented unit testing with MSTest and JSON import/export.

Complaint Management System

- Designed and implemented a layered front-end architecture using React and TypeScript, separating presentation (UI components), application/services, domain logic, and data access layers.
- Enforced clear dependency boundaries using service modules and interfaces to reduce coupling and improve maintainability using Google Firebase
- Managed state and side-effects through dedicated service layers to prevent business logic leakage into UI components.
- Wrote unit tests for domain and service logic using Jest, improving testability independent of UI concerns.

SKILLS

Languages: Python, JavaScript,
TypeScript, C#, PHP, SQL,
HTML, CSS

Frameworks: Flask, React,
Laravel, Node.js

Databases: PostgreSQL,
MySQL, SQL Server

Tools & Practices: Git, Azure,
REST APIs, Selenium, Unit
Testing, Agile/Scrum